

# Efficient Fault Diagnosis of Induction Motors Using Custom k-NN and Principal Component Analysis

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DS-323: Machine Learning for Data Science

Assignment-01

**Abstract**—This project investigates the application of machine learning for the automated diagnosis of faults in three-phase induction motors. By using a custom K-Nearest Neighbors (k-NN) algorithm that was made from scratch to detect motor faults across 14 simulated motor industrial conditions via binary classification. The study addresses the high dimensionality issues of signal data by incorporating Principal Component Analysis (PCA) to reduce 1000-sample features into 30 principal components which cover maximum variance of data. Furthermore, Principal Component Analysis (PCA) was applied to analyze the trade-off between dimensionality reduction and classification accuracy. Results indicate that the model achieves peak performance at an optimized K value of 7, gaining an accuracy of 97.98

**Index Terms**—Fault Diagnosis, Induction Motor, k-NN, Principal Component Analysis, Machine Learning

## I. PROBLEM STATEMENT

The goal is to perform binary classification on three-phase current signals to distinguish healthy motor operation from thirteen specific fault states that are labeled as unhealthy. We manage the complexity of high-dimensional raw data by implementing PCA and a custom k-NN algorithm from scratch, without the use of standard machine learning libraries. This method helps in predictive analysis in industrial fields

## II. DATASET DESCRIPTION

The dataset consists of instantaneous values of three-phase current (Current A, B, and C). •Health Conditions: Healthy, Inner-race/Outer-race bearing faults (0.7mm–1.7mm), and Broken Rotor Bar (BRB). • Current Loads: 100W, 200W, and 300W. • Data Structure: 39 datasets across 7 folders. Each raw file contains over 100,000 samples.

## III. DATASET PREPROCESSING

Raw data is partitioned into blocks of 1,000 samples per channel to capture complete signal cycles. Each 1,000-sample segment is assigned a "Healthy" or "Unhealthy" label to create a supervised binary classification dataset. Standard Scaling is applied to ensure that all signal variations have equal weight during the Euclidean distance calculation, preventing large magnitudes from biasing the model.

## IV. METHODOLOGY

The KNN algorithm was built from scratch to classify motor health by measuring the distance between data points using the Euclidean formula. It identifies whether a motor is healthy or

unhealthy based on the labels of its closest neighbors in the dataset.

## Testing and Validation

To ensure the model is reliable, we used hold-out testing and 10-fold cross-validation. This involves splitting the data multiple times to verify that the classifier works accurately on new, unseen signals.

## Data Compressing (PCA)

PCA was applied to shrink the 1,000-sample data blocks into 30 essential features. This makes the calculations much faster while still retaining 99.9% of the important information needed for a correct diagnosis.

## V. RESULTS AND DISCUSSION

*A. A. K-Value Optimization* The model was tested across multiple iterations to determine the most effective neighbor count for accurate binary classification. A K-value of 7 was identified as optimal, delivering the highest classification accuracy of 0.9798.

| Model Performance by K (Binary Classification) |          |           |        |             |             |          |
|------------------------------------------------|----------|-----------|--------|-------------|-------------|----------|
| K Value                                        | Accuracy | Precision | Recall | Sensitivity | Specificity | F1 Score |
| 1.0000                                         | 0.9796   | 0.5340    | 0.5392 | 0.5392      | 0.9895      | 0.5366   |
| 3.0000                                         | 0.9792   | 0.5258    | 0.5000 | 0.5000      | 0.9899      | 0.5126   |
| 5.0000                                         | 0.9789   | 0.5208    | 0.4902 | 0.4902      | 0.9899      | 0.5051   |
| 7.0000                                         | 0.9798   | 0.5417    | 0.5098 | 0.5098      | 0.9903      | 0.5253   |
| 11.0000                                        | 0.9779   | 0.4948    | 0.4706 | 0.4706      | 0.9892      | 0.4824   |
| 17.0000                                        | 0.9774   | 0.4845    | 0.4608 | 0.4608      | 0.9890      | 0.4724   |

Best K Value: 7, Accuracy: 0.9798

Fig. 1. KNN

*1) B. Hold out and K-fold using best k Value:* A stable 0.9797 cross-validation accuracy confirms the custom k-NN generalizes effectively and performs consistently across all unseen data segments.: The model achieved a 97.98% hold-out accuracy and high specificity, proving it is extremely reliable at identifying healthy motor states.

## C. PCA Impact on Results

PCA retained 99.94% variance with 30 components, ensuring 97.92% accuracy while significantly increasing processing speed.

## VI. ALGORITHM AND PSEUDOCODE

### A. Custom k-NN

Calculate the distance between a new motor reading and every known sample in the library.

Average metrics across all folds

Accuracy : 0.9797  
Precision : 0.5471  
Recall : 0.5790  
Sensitivity: 0.5790  
Specificity: 0.9888  
F1 Score : 0.5578  
TP : 20.3000  
TN : 1499.2000  
FP : 17.0000  
FN : 14.5000

Fig. 2. 10 K-folds Average

Hold-Out Metrics (PCA + k-NN):

Accuracy : 0.9792  
Precision : 0.5248  
Recall : 0.5196  
Sensitivity : 0.5196  
Specificity : 0.9895  
F1 Score : 0.5222  
TP : 53  
TN : 4503  
FP : 48  
FN : 49

Fig. 4. Hold-out with PCA

Results for Hold-Out Method with K=7:

Accuracy : 0.9798  
Precision : 0.5417  
Recall : 0.5098  
Sensitivity: 0.5098  
Specificity: 0.9903  
F1 Score : 0.5253  
TP : 52  
TN : 4507  
FP : 44  
FN : 50

Fig. 3. Hold-out

Average metrics across all folds:

Accuracy : 0.9371  
Precision : 0.2468  
Recall : 0.8643  
Sensitivity: 0.8643  
Specificity: 0.9388  
F1 Score : 0.3834  
TP : 21.2000  
TN : 996.2000  
FP : 64.9000  
FN : 3.4000

Average CV Accuracy (PCA + k-NN, K=7): 0.9371

Fig. 5. K-fold with PCA

Identify the K closest neighbors and assign the most common health label as the prediction.

### B. PCA Transformation

Center the signal data and identify the primary directions of maximum variance.

Project the 1,000-sample blocks onto the top 30 directions to create a compressed version.

Validation Process

Divide the dataset into training and testing portions to evaluate the model on unseen signals.

Perform 10-fold cross-validation by rotating the test data to ensure consistent accuracy.

## VII. FRAMEWORK

DataAcquisition → Preprocessing → Customk-NN Classifier → Hold-out → k-fold → Prediction. DataAcquisition →

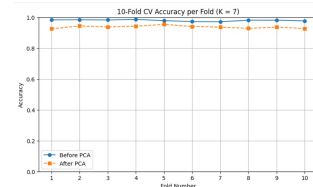


Fig. 6. PCA

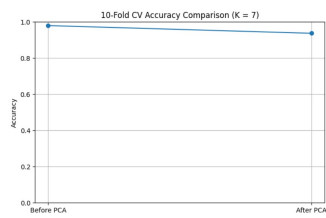


Fig. 7. PCA

Preprocessing → PCA → Customk-NN Classifier → Hold-out  
→ k-fold → Prediction.

#### REFERENCES

- [1] Machine Learning for Data Science, "DS-323 Assignment Guidelines," 2025.
- [2] Dataset Documentation for Three-Phase Current Signatures.